

Total No. of Questions : 8]

SEAT No. :

PD-4304

[Total No. of Pages : 2

[6403]-102

T.E. (E &TC)

EMBEDDED PROCESSORS

(2019 Pattern) (Semester - VI) (Elective- II) (304195)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q1 OR Q2, Q3 OR Q4, Q5 OR Q6, Q7 OR Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Draw an Interfacing diagram of GSM-UART module with LPC2148 and write an initialization program to send a message. [9]

b) State features of ADC in LPC2148, draw an interfacing diagram to display the sensed temp on LCD with initialization program. [9]

OR

Q2) a) Explain UART module of LPC2148. Write an embedded C program to transmit character 'A' to PC. [9]

b) Draw an interfacing of DHT 11 with LPC2148. Write algorithm to display temperature and humidity. [9]

Q3) a) Compare Cortex processors over ARM7 for embedded system design. Enlist the features and applications of Cortex processors. [9]

b) Draw and explain the memory map of Cortex M Processor [8]

OR

Q4) a) State various Features of STM32F4xxx, Comment on Peripheral Clock [9]

b) Explain the programmer model-Register in Cortex M Processor. [8]

P.T.O.

- Q5) a)** Explain timer structure of STM32F4XX. Which are the SFRs associated with timers? Write Embedded C program to generate delay of 5 msec.[9]
- b)** Draw an interfacing diagram to interface LDR & MQ3 sensor with STM32F4xx microcontroller and write algorithm to display the light & the Gas percentage parameter on LCD. [9]

OR

- Q6) a)** What are the features of GPIO of STM32F4XX? Write a note on different types of GPIO registers available in STM32F4xx. [9]
- b)** Write a program to generate 100 ms time delay using timer. Assume external clock frequency of 16 MHz. [9]
- Q7) a)** State features of CAN Bus with CAN Bus frame. [9]
- b)** Draw an interfacing diagram of STM32F4xx with accelerometer MPU 6050 with flowchart to display the parameters. [8]

OR

- Q8) a)** Write detailed note on PWM. With interfacing diagram, show speed and direction of DC Motor can be changed using PWM in STM32F407XX. [9]
- b)** State features of ultrasonic sensor-HC-SRO4, Draw an interfacing diagram. [8]

